

Certified analytical bounds for hypersonic reachable sets

A two-page summary of a preliminary PhD research proposal

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The question. What I want is a rigorous and quantitative guarantee for atmospheric entry, and the mechanism I expect to make it tractable — the concentration of the aerodynamic forcing into a thin layer — is what I intend to contribute. An entry vehicle is steered by its bank angle, which turns the direction of its lift without changing the magnitude, and over a single descent it may enter the dense part of the atmosphere several times, rising between entries. The design question is the *reachable set*: the states, or the landing sites, consistent with the admissible controls and with an atmosphere known only within bounds. I propose to study the structure of that set in the limit in which the atmosphere is thin, by the methods of nonlinear analysis rather than by simulation.

What makes the limit singular. Let ε denote the ratio of the atmosphere’s scale height to the planetary radius; for the Earth $\varepsilon \approx 1.3 \times 10^{-3}$. In this limit the aerodynamic force is appreciable only in a thin band of altitudes. A vehicle crossing the band spends a time there that vanishes with ε . Whether it leaves with its flight-path angle changed by an amount that does not, and in which variables, is the first question of the analysis rather than one of its assumptions: a force of order one acting for a time of order ε delivers a vanishing impulse, so an effect that survives has to be weighed against an entry angle itself small with ε . That a crossing is brief and decisive at once, in whichever variables make it so, is the feature I want to analyse. It is also what makes the standard tools awkward. A limit that weighs an interval of time by its length records the duration of a crossing, which tends to zero, rather than the change it produces, which does not. An averaging theorem [1] returns an order of approximation on a bounded interval, where a reachable set requires an inclusion holding for every admissible control. And the natural formulation is a differential inclusion [2] rather than a single controlled trajectory, since the admissible controls and the admissible atmospheres are both to be quantified over.

The model I propose to study. I propose to carry out the analysis first on a reduced model that keeps the feature above and discards everything else: a particle in a potential with one soft wall, the control entering as the strength of that wall,

$$\dot{z} = v, \quad \dot{v} = c \rho_\varepsilon(z) - \kappa \rho_\varepsilon(z) v - 1, \quad \dot{s} = F(s, c) + \rho_\varepsilon(z) \Gamma(s, c),$$

with z an altitude, v its rate, s the slow variables of the descent, $\rho_\varepsilon(z) = \rho(z/\varepsilon)$ a positive decreasing density profile read on the scale ε , c the control, and κ the drag; both aerodynamic terms carry the factor $\rho_\varepsilon(z)$ explicitly, since neither lift nor drag acts outside the air. One altitude here is determined rather than chosen: the level at which $c \rho_\varepsilon = 1$ and the lift balances gravity, which is the minimum of the potential and the glide. The reduced model has to be derived rather than posited, and that is where I would start: which distinguished scaling makes a crossing survive the limit as a transition of its own, how the crossings should then be delimited, for which I want something intrinsic to the motion rather than a chosen reference altitude, and which features of the full entry equations [3] the reduction may drop. In the full problem it is the ratio $\gamma_e/\sqrt{\varepsilon}$, formed from the entry flight-path angle, that governs how a crossing resolves, as in the classical Chapman variables [4].

The structure I expect, and where it comes from. The shape I expect is the one familiar from concentration compactness: a skeleton carrying the slow descent, one profile for each crossing of the band, and a remainder. In joint work with E. Kirr on multi-peak ground states of the nonlinear Schrödinger equation we used the concentration compactness principle [5] to decompose a family of solutions into finitely many profiles at distinct centres, each resolved at its own scale [6, 7], and a Lyapunov–Schmidt reduction to turn the interaction between the profiles into a finite-dimensional algebraic system for their positions. The analogy I want to test is that a crossing of the band plays the part of a peak: it has its own scale, which here is the width of the band, and its own centre, which here is the time at which it occurs. Two differences are the substance of the proposal. The objects are trajectories of a controlled system rather than critical points of a functional, and what must be bounded is a whole family of them rather than a single sequence. And the decomposition would have to be quantitative: the classical argument produces a finite profile count by contradiction, whereas a reachable-set bound needs that count with a constant, in terms of the data.

What would have to be established. Four problems, in the order I would attack them.

1. **A decomposition, with a constant.** That every sequence of admissible trajectories splits, along a subsequence, into a skeleton, finitely many profiles obtained by rescaling the crossings to the width of the band, and a remainder small in a stated norm — and that the number of profiles is bounded in advance by the data rather than by contradiction. This requires first specifying the space the trajectories live in, the group acting on it, and the sense in which the remainder is small; none of the three is settled.
2. **A limit object, and an inclusion.** That the reachable sets of the family converge, in a sense to be chosen, to that of a limit object in which each crossing has become an instantaneous transition. The transition should be

set-valued, the control being free during a crossing, and the inclusion one-sided and explicit, an outer bound being what a design question needs.

3. **The companion at the level of measures.** The occupation-measure formulation of Lasserre, Henrion and collaborators [8] renders the dynamics a linear constraint, and is where I expect the limit to be cleanest to state; a crossing of no duration carries no ordinary occupation mass, so a separate transition measure on the graph of the limiting map may have to be introduced alongside. Whether it is better taken there or at the level of trajectories, and whether the two agree, I do not know.
4. **The full entry problem.** The attitude and structural motions do not concentrate and have to be handled separately; the atmosphere is uncertain and time-varying; and the guidance is issued at finite intervals rather than continuously. Each is a question about which hypotheses the analysis can honestly carry.

What I am asking. I am looking for an advisor in nonlinear analysis for whom this is a congenial problem. My training for it is the NLS work above: the decomposition I expect here is one I have already worked with, and what is new is that the objects are trajectories of a controlled system rather than critical points of a functional. I have a draft route to the inclusion of the second problem for the reduced model, resting on elementary tools, and I would like to take it to a board and have it examined. The entry problem is the objective itself and not an application to be added once the mathematics is finished, so the fit I am looking for is someone willing to work at that interface.

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